Results in Engineering 29 (2026) 109865

Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering

Research paper

Explainable hybrid deep learning framework for cryptocurrency price forecasting using CNN-BiLSTM-TCN and SHAP analysis

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ARTICLE INFO

Keywords:
Cryptocurrency forecasting
CNN
BiLSTM
TCN
Hybrid deep learning
Time-series prediction
SHAP
Explainable AI (XAI)

ABSTRACT

Cryptocurrency markets are highly volatile, non-linear, and have complex time dependencies which in turn makes accurate forecasting a difficult problem. This paper presents a new hybrid deep learning architecture that uses Convolutional Neural Network (CNN), Bidirectional Long-Short-Term Memory (Bi-LSTM), and Temporal Convolutional Network (TCN) in order to combine the benefits of a recurrent model and a convolutional method. The model is secondary trained a multi-feature cryptocurrency dataset which does not obey log returns and incorporates temporal dependencies and augmentation on generalization by different architectural connections. To add transparency and interpretability to the predictions, we add SHAP (SHapley Additive exPlanations) analysis to the hybrid model and can attribution of features to time steps. Our solution gives the user an insight into how and why their models may be performing as they do with regard to showing which parts of their time series are most relevant and how they interact with each other, rather than focusing only on empirical results. Experimental findings indicate that the proposed hybrid model is much better in comparison with individual deep learning models in terms of RMSE, MAE, R-Square, and SHAP analysis proves that the model makes reasonable decisions that are consistent and explainable. The work offers an encouraging approach to explainable financial prediction: it has a high predictive performance and explains how predictions are made. It is possible to apply the approach to other fields of high-frequency time series outside the sphere of cryptocurrencies.

1. Introduction

Over the past decade, cryptocurrencies, especially the Bitcoin, have become an outstanding innovation in the modern finance [1]. Since its inception by Satoshi Nakamoto in 2009, Bitcoin's decentralization structure and the peer-to-peer transactions have become a big challenge for traditional monetary systems through enabling transactions without central intermediaries . Moreover, Bitcoin's rapid growth and its turbulent price fluctuations, more particularly, in its prosperous era during the years 2017 and 2021, all contributed to turn Bitcoin into both a digital shift symbol and a source of high-stakes volatilit[2].

In addition to long-term growth phases, cryptocurrency markets frequently experience sudden price swings, flash crashes, and sharp recoveries driven by regulatory announcements, macroeconomic events, and speculative trading. These abrupt fluctuations significantly increase uncertainty and make accurate forecasting particularly challenging.

The digitalization of global economy has become a sought-after trend because of the amplification of the process. According to recent reports, the volume of digital economy will achieve 23 trillion USD by 2025. Which represent 25% of global GDP [3]. In the realm of this fast expanding system, the prediction of cryptocurrencies' prices has become a fruitful ground for investors, monetary institutions, and policy makers who seek accurate forecasting that ensures a better management of risks, foster profitability, and set efficient rules and regulations that fit the new developed ecosystem.

Despite the spread-out of Bitcoin, its volatility stands as an obstacle to a stable investment and macro-financial planning. Traditional economic models such as ARIMA [4] and GARCH [5,6] postulate linearity and stationary of data, and natural distribution of errors. These assumptions seldom come true in markets that know non-linear dynamics, heavy tails distributions, and sudden system changes. As a result, the performance of these models is usually weak when facing sudden reflections or non-stable behavior, which creates a methodological gap in trustworthy Bitcoin prices predictions.

To overcome these restrictions, researchers have increasingly been adopting machine-learning (ML) and deep-learning (DL) methods that are potentially able extract non-linear relationships and hidden temporal dependencies without strict statistical assumptions. Initial machine-learning methods such as Multilayer Perceptrons (MLP) and Support Vector Machines (SVM) [7] have contributed to the amelioration of

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https://doi.org/10.1016/j.rineng.2026.109865
Received 31 December 2025; Received in revised form 17 February 2026; Accepted 1 March 2026
Available online 8 March 2026
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flexibility, but they failed to tackle over-learning problems and temporal relations understanding.

The emergence of Recurrent Neural Networks (RNNs)[8]–notably Long Short-Term Memory (LSTM)[9] and Gated Recurrent Units (GRU)–made it possible for models to capture sequential dependencies within volatile financial series[10] .based on these facts, hybrid deep architectures combining CNN, LSTM, and attention mechanisms have achieved a remarkable accuracy in cryptocurrency trends prediction[11].

More recently, Transformer-based and sentiment-integrated forecasting frameworks have demonstrated strong performance in modeling complex temporal patterns. However, these approaches often require large-scale datasets and high computational resources. In contrast, the proposed CNN-BiLSTM-TCN framework provides a computationally efficient alternative while still capturing multi-scale temporal dependencies, making it suitable for practical financial forecasting environments with limited data and resources.

This study presents an interpretable hybrid deep-learning framework that merges Convolutional Neural Networks (CNN), Bidirectional LSTM (BiLSTM), and Temporal Convolutional Networks (TCN) to achieve an effective and trustworthy cryptocurrency prediction.

The integration of these three components is theoretically motivated by their complementary strengths in modeling multi-scale temporal structures within financial time series.

- **The CNN** component captures short-term spatial features and localized volatility by detecting local patterns and rapid fluctuations in price movements.
- **The BiLSTM** module learns bidirectional temporal dependencies within sequential market data, allowing the model to understand how past and future contexts jointly influence current price behavior.
- **The TCN** enables the expansion of receptive fields through dilated causal convolutions, allowing efficient modeling of long-range dependencies while maintaining stable gradient flow.

These components are combined with **SHAP-based explainability**, to ensure transparent feature attribution and model accountability.

In financial forecasting environments, where predictive models can influence investment strategies and risk management decisions, interpretability is critically important. In this study, SHAP is integrated to provide feature-level explanations of the hybrid model's predictions across time steps, allowing analysts to understand which price signals most strongly influence the forecast. This improves model transparency and supports more informed and trustworthy decision-making in practical financial applications.

The objectives of the study in hand are to:

1. Formulate a unified CNN-BiLSTM-TCN hybrid architecture that aims at modeling multi-scale temporal dependencies in Bitcoin price dynamics.
2. Assess its predictive performance against standalone DL baselines using MAE, MSE, RMSE, and R^2 metrics.
3. Anchor SHAP interpretability to improve transparency, reliability, and ethical deployment in financial forecasting systems

This work contributes to the next generation of trustworthy and interpretable AI for cryptocurrency price prediction by bridging the gap between high predictive accuracy and practical decision support through the integration of hybrid deep learning and explainable artificial intelligence.

2. Related works

This work contributes to the next generation of trustworthy, interpretable AI for cryptocurrency price prediction–bridging the gap between computational accuracy and responsible decision-making through combining predictive precision with post-hoc explainability.

The following section provides a general overview on the current research papers related to cryptocurrency prices prediction focusing on the use of machine learning and deep learning models. The researchers applied a set of algorithms to predict cryptocurrency prices, such as LSTM model and other deep learning techniques. Lately, deep learning methods have become increasingly favored to predict temporal series, especially in analyzing cryptocurrency markets volatility.

One of these methods relies on LSTM model when predicting cryptocurrency prices. Boongasame and Songram [12] have compared LSTM model efficiency to other statistical methods and traditional analysis techniques such as SMA (Simple Moving Average), and WMA (Weighted Moving Average), and EMA (Exponential Moving Average). Mean Absolute Percentage Error (MAPE) is employed in this study as a main evaluation standard. The results have shown that LSTM model, particularly when optimized by the right time-window features, has achieved a high predictive accuracy and surpassed the traditional methods.

Fleisher et al. [13] have developed a prediction system based on LSTM, and its performance has been compared to ARIMA classical model. Their findings showed that the LSTM model has achieved lower Root Mean Squared Error (RMSE) proving an effective ability to detect nonlinear changes in cryptocurrency prices. Similarly, Patel et al. [14] have examined various deep learning and assembly techniques, including LSTM, GRU, and Random Forests, in order to accurately predict Bitcoin prices. Their results have demonstrated that hybrid neural networks significantly outperformed the independent models thanks to their ability to model both sequential and structural series.

Tripathy et al. [15] evaluated several algorithms–ARIMA, LSTM, and Facebook's FB-Prophet–for Bitcoin price forecasting, and have come to the conclusion that the latter (FB-Prophet) provides a greater ability to adapt in the volatile market. Meanwhile, Wu et al. [16] introduced Transformer-based neural architectures to predict temporal series, to show that self-attention mechanisms can effectively represent long term correlations without facing the problem of fading gradients. This invention is a crucial step towards deep learning models which have greater potentials to predict cryptocurrency prices.

McNally et al. [17] have compared Recurrent Neural Networks (RNN), LSTM, and ARIMA for Bitcoin prices prediction. The results have shown that LSTM achieved the highest accuracy classification (52%) and the lowest RMSE (8%), this result confirms its prediction supremacy compared to traditional economic methods. Wu et al. [18] have expanded this research line through combining autoregressive components with LSTM networks in a hybrid LSTM-AR(2) model, which led to an accuracy improvement in short-term currency prices predictions.

Li and Dai [19] have designed a CNN-LSTM hybrid model that extracts spatial short-term treats using CNN layers, and detects long-term temporal correlations using LSTM layers. Their model has achieved considerable values; MAE = 209.89, RMSE = 258.31, and MAPE = 2.35, outperforming the foundational models. By the same token, Ramos-Pérez et al. [20] have used LSTM and GRU networks to predict the prices of Bitcoin, Ethereum, and Litecoin, remarkable performance was achieved compared to traditional models. In a later study, the same researchers have presented a Multi-Transformer framework that extends the transformer structure to predict the S&P index volatility, showing its different uses in various monetary tools.

Kanaparthi [21] has assessed the LSTM networks strength against Gaussian noise, showing that LSTM has outperformed the ARIMA(2,1,2) model even when the date were unstable due to random Inconstancies. Labbaf and Manthouri [22] have presented an enhanced BiLSTM model combined with Performer attention mechanism, which allows an effective detection of long-term reliances and a high accuracy of prediction through datasets of Bitcoin, Ethereum, and Litecoin. . Jin and Li [23] have developed a VMD-AGRU-RESVMD-LSTM hybrid structure that encompasses variational mode decomposition and residual correction providing a great improvement of prediction accuracy without compromising computational competency.

Chen [24] has analyzed the Random Forest Regression (RFR) and LSTM predictive capabilities through 47 explanatory financial and macroeconomic variables. The study has shown that the historical

Bitcoin prices, main shares indicators, and Ethereum prices were the most influential factors, the fact that emphasizes the interrelation between traditional and crypto-asset markets. Ladhari and Boubaker [25] have investigated the high frequency prediction using two hybrid models –LSTM-Attention and ANN-LSTM– which are trained on more than 50.000 hourly samples. Their results have proven that the LSTM model, reinforced with attention, has achieved a higher predictive accuracy thanks to its ability to dynamically prioritize the key temporal features.

Frohmann et al. [26] have adopted a hybrid approach based on BERT through combining sentiment analysis with textual quantitative data. The results have shown that incorporating sentiment information, particularly through favoring the tweets based on the impact of their owners, has noticeably contributed to the enhancement of the predictive performance (MAE = 2.67, RMSE = 3.28). it has also been proven that the multidimensional inputs, including sentimental and behavioral indicators, have played an important role in modelling and predicting the cryptocurrency prices.

Bourday et al. (2024) [27] have broadened this approach through conducting a comparative and comprehensive analysis of hybrid structures that rely on Transformer-based models–including Transformer-XGBoost, Transformer-ANN, Transformer-LSTM, and Transformer-SVR– in order to predict Bitcoin prices in short-term and mid-term. Researchers relied on historical data from 2016 to 2023 to demonstrate that the Transformer-XGBoost model has achieved the highest accuracy (MAE = 0.011, RMSE = 0.018) outperforming all main deep-learning models. This study has set Transformer based hybrid models as strong and expendable frameworks for modeling complex financial systems.

Based on this work, Bourday et al. (2025) [28] have presented a transfer-learning-enhanced hybrid model reinforced with transformed learning, which combines GRU, BiLSTM, and Multi-Head Attention mechanisms with sentimental data from twitter. Using a two-stage training strategy–pretraining (2017–2019) followed by fine-tuning (2021–2023)– this model has achieved RMSE = 679.61 and MAE = 452.95, which shows that sentiment-informed transfer learning highly improves both generalization and predictive strength.

In addition to these advances, Ahmed et al. [29] have suggested a Hybrid Convolutional Recurrent Model (HCRM) that combines a one-dimensional CNN for feature extraction with an LSTM for temporal regression, and it has achieved an outstanding accuracy in predicting Bitcoin prices. Similarly, Bouslimi et al. [30] have confirmed the performance of LSTM networks using high frequency data of digital currencies during COVID-19 pandemic, showing that models edited with 100 hidden units and 200 epochs achieve a typical accuracy under extreme volatility conditions.

Recent methodological advancements have highlighted the importance of capturing multi-scale temporal dependencies and stochastic nonlinearities in complex dynamic systems. Zhang [31] have introduced MamNet, a hybrid architecture that integrates Mamba-based state-space modeling with Fourier-based spectral decomposition to simultaneously capture long-term dependencies and periodic frequency patterns. Complementing this framework, Liu and Ding [32] have proposed a Newton-based iterative estimation algorithm for bilinear stochastic systems, employing a bilinear state observer to ensure a solid parameter convergence under nonlinear uncertainty. Beyond temporal dynamics, recent research has demonstrated that integrating Prospect Theory with evolutionary game dynamics effectively captures behavioral heterogeneity–such as loss aversion and asymmetric risk perception–within decentralized trading environments [33,34].

Collectively, these frameworks provide strong theoretical justification for the deployment of hybrid deep learning architectures in cryptocurrency forecasting; as such models must approximate the complex interplay between multi-scale temporal structures and non-rational market sentiment. Furthermore, anchoring data-driven predictions in game-theoretic and behavioral principles enhances interpretability, enabling tools such as SHAP to attribute volatility not merely to statistical patterns but to underlying strategic interactions and behavioral biases.

3. Methodology

A hybrid approach (including data preprocessing, the construction of deep learning models, and their hybridization) with an analysis of interpretability was developed to solve the problem of volatility and complexity of the time series data of cryptocurrencies.

The overall pipeline consists of data normalization, sequence generation, individual model training (CNN, BiLSTM, and TCN), feature fusion through hybridization, and SHAP-based interpretability analysis.

The training configuration was selected based on preliminary experiments to ensure stable convergence and prevent overfitting. Across all models, training was conducted for 30 epochs, which was sufficient for both training and validation losses to stabilize. A batch size of 32 was adopted to balance computational efficiency and gradient stability, which is commonly used in deep learning-based time-series forecasting tasks.

The steps in the pipeline follow normalization of the multivariate features (Open, High, Low, Close) to render all variables on similar scale without use of log returns or denoising, hence resulting in retaining the raw temporal patterns. Sliding window technique was then applied to convert the sequential data stream into fixed length input stream.

Within three state of the art architectures, Convolutional Neural Network (CNN), Bidirectional LSTM (BiLSTM), and Temporal Convolutional Networks (TCN) were implemented in turn and tested.

A hybrid model was then constructed by extracting latent feature representations from each architecture and concatenating them into a unified representation before passing them through dense layers for final prediction.

In the hybrid framework, each branch learns complementary temporal information: CNN captures localized patterns, BiLSTM models bidirectional temporal dependencies, and TCN captures long-range sequential relationships. The learned feature vectors from these branches are concatenated to form a high-dimensional latent representation, enabling the model to integrate multi-scale information for improved prediction accuracy.

SHAP (SHapley Additive exPlanations) was also used to visualize and interpret the influence of each of the input features on the time steps bringing transparency and explainability to decision-making processes of the model,as depicted in Fig. 1.

3.1. Dataset description

This project involves the use of past Bitcoin (BTC) prices downloaded to the Kaggle dataset Cryptocurrency Price History by Sudalai Rajkumar. This data set offers time-series daily records starting April 2013 up to date and end 2021-07-07 and all crucial indicators about the market like:

- Date : Day by day time stamp
- Open : Opening price of the day
- Highest : Maximum price of the day
- Low : Low price of the day
- The close of the day : Price at the end of the day
- Volume : Daily transaction volume (put at zero in case of early entries)
- Market Cap : The sum of market valuation of Bitcoin on that day

In this work, Bitcoin only was chosen to keep a focused model where there is less noise due to multi-coin dynamics. Open, High, Low, and Close prices are the features used in training the model.

The Volume feature had been omitted because there were zeros in the old records which could introduce noise and instability during training. Although missing value handling techniques such as interpolation or imputation could be applied, they may introduce artificial patterns in financial time-series data. Therefore, this study prioritizes data reliability by focusing on OHLC features, while leaving the integration of volume-based indicators as future work.

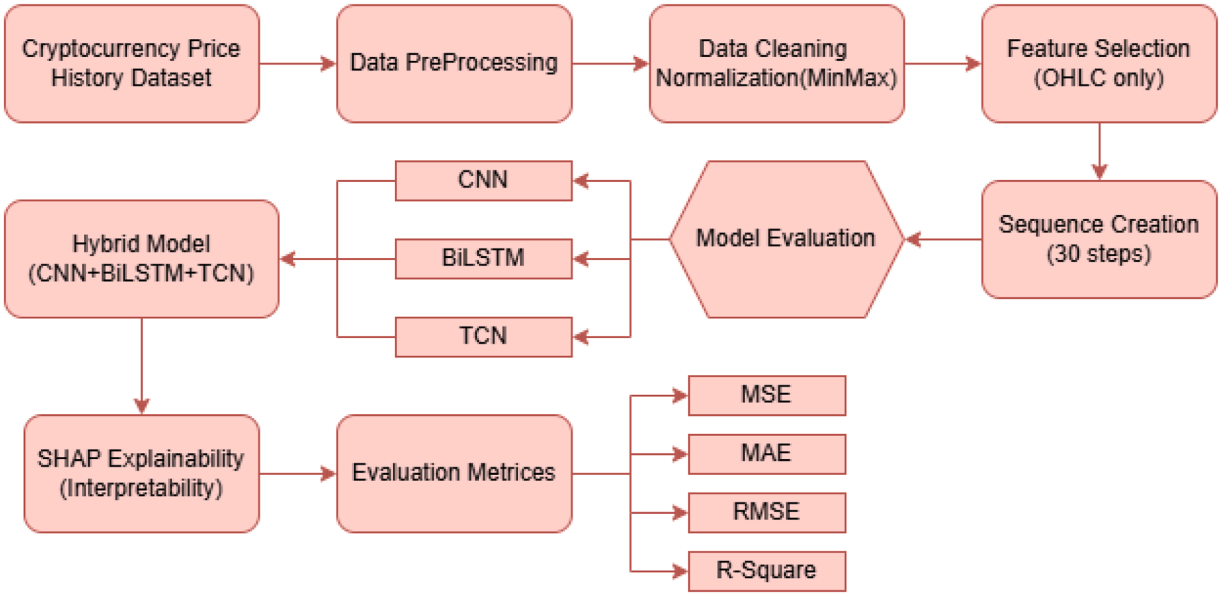


Fig. 1. Bitcoin price prediction pipeline based on a hybrid CNN-BiLSTM-TCN architecture with SHAP-based interpretability.

Missing value treatment, Min-Max normalization, and preparation of sequences to operate deep learning models were considered data preprocessing. The end result after the preprocessing are 2990 daily records remodeled into the time sequence with 30 time points with 4 features.

3.2. Data preprocessing

The preprocessing stage was initiated by taking out the historical data of the Bitcoin price (original dataset). Pandas were used to parse the dataset and the date column was converted to proper date time so that it was sorted chronologically. To ensure that the regulatory approach of the model reflects core market behaviors, features such as volume and market cap were not chosen as a lot of data were either missing or of the non-informative nature.

Data was normalized using Min-Max scaling before feeding it into the deep learning models. Min-Max normalization was selected over alternatives such as Z-score normalization because it preserves the original distribution shape while scaling features into a fixed range [0,1], which improves gradient stability and accelerates convergence in deep neural networks. This is particularly important in financial time-series forecasting where features may have different scales.

The normalization formula is:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

Where: The initial value of one of the features is x,$x_{\min}$ and $x_{\max}$ are the lowest and the biggest values of the feature respectively. This normalization process has the advantage of ensuring that every feature of the input data makes equal contribution when training the model and preventing dominance of any particular feature. This normalized dataset was then visualized in order to verify effectiveness of scaling, and processed further in order to generate moving sequences of 30 consecutive days of data as input and the next day price as the target to run time-series forecasting.

3.3. Sliding window sequence generation

In order to represent temporal complexity of the price movement of Bitcoin, a sliding window method was used where normalized data was categorized into the supervised format of learning which can be applied to a time series forecasting algorithm. The window size was set to 30 days, this is, each input sample, that is fed into the model will be 30 consecutive days of price features namely: 'Open', 'High', 'Low' and 'Close'.

The window size of 30 days was selected to provide enough recent historical context for learning short-term price movements while keeping the model training manageable.

The matching target price is 'Open' of the 31 st price so that the model learns the past trends and is able to predict the action of the market the following day. The creation of this transformation was performed with a custom create_sequences() method and computes it by iterating through the normalized dataset, segmenting each into overlapping subsequences of shape (30, 4) and saving the following target. Two NumPy arrays are the products of this process: X_seq: Input attributes of shape (samples, 30, 4), y_seq: Targets of shape (samples,). With the purpose of checking accuracy, one of the generated input sequences was plotted in order to conduct the visual inspection. The plot of every line shows how the temporal changes of a discrete value of a price feature proceed over the time, and the structure of the learning patterns that the model will be exploiting during a training process is also validated. Given a multivariate time series:

$$D = \{x_1, x_2, \ldots, x_T\}, \text{ where } x_T \in R^F \tag{2}$$

D: The complete dataset of size T. x_t: A feature vector at time t with F features (here, F = 4: Open, High, Low, Close). Thus, the total number of sequences generated is:

$$N = T - w \tag{3}$$

3.4. Data splitting

To feed the data to supervised learning, the preprocessed and windows sequences would be subdivided into training and testing sets with ratio of 80:20. In this case, 80% of information was used to train the model and the rest of 20% was reserved as a method of evaluation. Noteworthily, the option shuffle = False was set to avoid loss of temporal structure of the data. In time series forecasting problems, it is important to ensure chronological order so as to avoid data leakage, which would otherwise cause overestimation of model performance and that would be unrealistic.

3.5. CNN for time series forecasting

A Convolutional Neural Network (CNN) model was developed to forecast Bitcoin prices by capturing local temporal dependencies within the time series.This model processes four input variables –Open, High, Low, and Close prices– using one-dimensional convolutional layers that extract short-term spatial features. Each convolutional layer is followed by ReLU activation and max-pooling so as to boost nonlinearity and reduce dimensionality. In order to prevent overfitting, the flattened output is passed through dense layers for regression, with a dropout rate of 0.2. Adam, with Mean Squared Error (MSE), as the loss function, is used to optimize the network. Training was conducted for 30 epochs with a batch size of 32 and an 80:20 train-test split.

The model has shown a smooth confluence and low error metrics, showing a great ability of generalization. On the whole, the CNN model has achieved a high efficiency in detecting price movement patterns and it has provided trustworthy short-term predictions of Bitcoin price dynamics.

CNN is particularly effective in capturing short-term volatility patterns and local price movements, making it suitable for identifying micro-trends in cryptocurrency markets.

3.6. Bidirectional LSTM (BiLSTM) based forecasting

In the second experimentation step, the BiLSTM model was created to address the problem of the time series and capture the relationship between the past and the future.

The architecture of the model was: A Bidirectional LSTM layer with 64 units, which can take into account both the backward and the forward information to improve the contextual perception. It is followed by a Dropout layer (rate = 0.2) to avoid overfitting. ReLU-activated Dense layer with 32 units to transform the features. One final output unit Dense layer to estimate the price of the next time step. The model was trained with Adam optimizer and mean squared error (MSE) via the mean. The training was done on 30 epochs with batch size of 32 and evaluation on held-out test set.

The bidirectional structure enables the model to learn relationships from both past and future contexts within the input window, enhancing its ability to model sequential financial behavior.

3.7. Temporal convolutional network (TCN) based forecasting

A Temporal Convolutional Network (TCN) has been placed so that longterm temporal trends could be better modelled and sequence modelling in time series forecasting could be enhanced. TCN (unlike recurrent structures LSTM, BiLSTM) uses causal and dilated convolutions and can enable parallel processing and a longer effective memory. TCNbased model has the following layers:

- Input layer: Shape (4 features, 30 time steps) can be presented at the input.
- TCN layer: 64 filters, kernel size = 3 , 0.1 dropout of regularization return_sequences = False in case a single prediction should come out
- The last dense(1) layer as the price forecasting layer.

Model is fully assembled with Adam optimiser and mean squared error (MSE) loss.To prevent over fitting we implement EarlyStopping callback that stops the model after a number of given epochs and monitors the validation loss of 5 epochs. The training of the model went forth on the split of training data in preprocessed time series, 30 epochs, a batch size of 32.

TCN offers advantages over recurrent models by enabling parallel computation, stable gradients, and efficient memory usage while maintaining temporal causality.

3.8. Hybrid deep learning model: CNN + BiLSTM + TCN

A combination of Convolutional Neural Network (CNN),Bidirectional LSTM (BiLSTM), and Temporal Convolutional Network (TCN), which have never been used together before in cryptocurrency price prediction, was created in this study to enhance the accuracy of the cryptocurrency price forecast and the strength of its results. In terms of inputs, the model expects a three-dimensional tensor with the (samples, 30, 4) shape, which is a sliding window of 30 time steps and 4 normalized price characteristics, Open, High, Low, and Close. In order to get a variable time pattern the input is sent in three parallel lines:

CNN branch Captures localized spatial and short-term temporal dependencies by applying onedimensional convolutional filters that detect hierarchical patterns across the input sequence. The core update mechanism is expressed as:

$$y_t = \sigma\left(\sum_{k=0}^{K-1} W_k x_{t-k} + b\right) \tag{4}$$

where x_t denotes the input at time step t, W_k represents the convolutional kernel weights of size K, b is the bias term, and $\sigma(\cdot)$ is a nonlinear activation function (e.g., ReLU).This enables the CNN to efficiently extract short-term correlations and trend indicators within the cryptocurrency price data.

BiLSTM branch processes sequences in both forward and backward directions to preserve past and future context. The hidden representation at time t is given by:

$$h_t = \overrightarrow{h_t} \oplus \overleftarrow{h_t} \tag{5}$$

where $\bigoplus$ denotes concatenation of the forward and backward hidden states. This allows richer temporal understanding than unidirectional models.

TCN branch uses dilated causal convolutions to model long-range dependencies with high computational efficiency. A dilated convolution at time t is expressed as:

$$y(t) = \sum_{k=0}^{k-1} f(k) \cdot x(t - d \cdot k) \tag{6}$$

where K is the filter size and d is the dilation factor. TCN offers parallel processing, stable gradients, and better memory access than recurrent models. The state outputs of the three branches are then joined together and a unified 256 dimensional latent representation is passed through two fully connected layers with ReLU activation which introduces nonlinearity as well as abstraction.

The latent feature vectors obtained from CNN, BiLSTM, and TCN branches are concatenated along the feature dimension to form a unified representation before being passed into fully connected layers for final prediction. No explicit weighting scheme was applied; instead, the network learns optimal feature importance automatically during training.

Lastly, a dense layer whose output is linear gives the prediction of the target value of the next time step (i.e., the 'Open' price).

3.9. Explainability via SHAP for interpreting hybrid forecasting model

SHapley Additive exPlanations (SHAP) were utilised in order to interpret the inner mechanics of the hybrid deep learning model (CNN + BiLSTM + TCN). SHAP is a common framework which comes up with cooperative game theory that gives a value to each feature on a specific prediction importance. In detail, the SHAP values are obtained by using the concept of Shapley values, which is calculated as:

$$\theta_i = \sum_{S \subseteq N\{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!}[f(S \cup \{i\}) - f(S)] \tag{7}$$

where ϕi represents the SHAP value of the feature i, N represents the entire set of input features and S is a subset that does not include i .

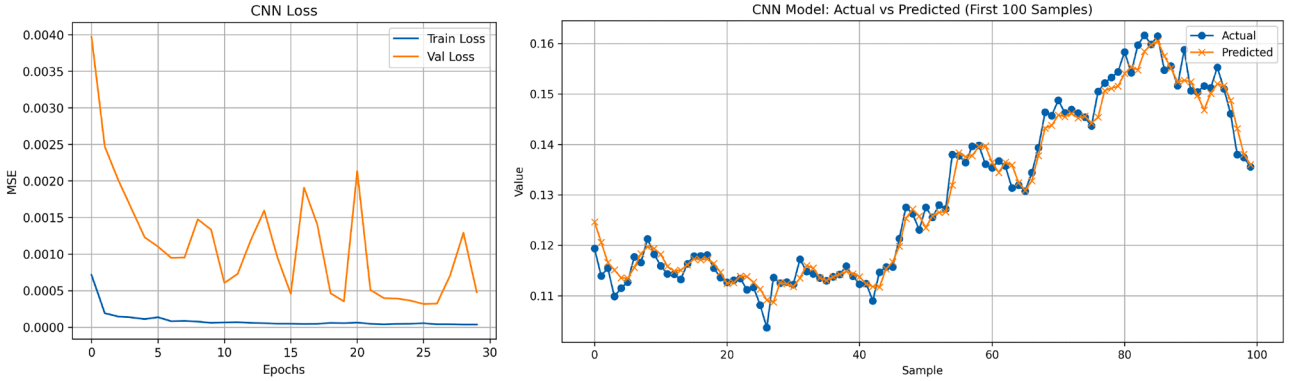


Fig. 2. Actual versus predicted Bitcoin prices using the CNN model: (a) training and validation loss convergence, and (b) comparison between predicted and true prices.

The formula is the mean marginal contribution of feature i considering all the subsets of features. The KernelExplainer of the SHAP library was applied in this experiment, and it is model-agnostic and estimating the Shapley values by local perturbations similar to LIME. The hybrid model requires input shape, as it is designed to handle it. Each sample of the test was remodelled to a 2D flat vector $(30, 4)$ and such a prediction wrapper function was established to map it back to model-compatible input 50 Samples out of a background dataset were chosen so that SHAP kernel approximation could be performed, and SHAP-values could be calculated on 10 test samples. The resulting SHAP array was a shape (10,120) and remodeled into a 3D tensor of (10,30,4) Ten to three zero four to synchronize with time structure. The average absolute SHAP value was averaged over all samples and points in time to measure the global feature importance:

$$\text{MeanImportance}_{\,J} = \frac{1}{N.T}\sum_{i=1}^{N}\sum_{t=1}^{T}\left|\Phi_i, t, j\right| \tag{8}$$

where $j \in \{1, 2, 3, 4\}$ is the four features Open, High, Low and Close. An importance plot was plotted as a bar-plot; this plot shows which of the components of the price equation has in the greatest extent impacted the predictions of the hybrid form. Also, the complete SHAP tensor was converted into a 2D matrix of shape $(300, 4)$. Calculate $(300, 4)$ and plotting it with SHAP paraphrase plots (violin/dot/bar). The images indicate the distribution and the size of SHAP values over the number of characteristics, marking the manner in which these characteristics contribute to prediction variance at various steps in time. The above SHAP-related interpretability framework does not only make the hybrid deep learning model more transparent but also makes it trustworthy and accountable, something that is significant in high-stakes areas such as financial forecasting.

3.10. Model evaluation metrics

In order to numerically analyze how the forecasting models (CNN, BiLSTM, TCN and Hybrid) serve, four popular regression measures were adopted, namely: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2 Score).

1. **Mean Absolute Error (MAE)** measures the average absolute difference between predicted and actual values. It is less sensitive to outliers and provides a clear understanding of average prediction error:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{9}$$

2. **Mean Squared Error (MSE)** evaluates the average squared differences between predictions and actual values. It penalizes larger errors more severely than MAE:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 \tag{10}$$

3. **Root Mean Squared Error (RMSE)** is the square root of MSE and expresses the error in the same unit as the target variable, which makes it easier to interpret:

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{11}$$

4. R^2 **Score** measures how well the model explains the variance in the actual data. An R^2 value close to 1 indicates that the model can explain most of the variability in the target variable:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}_i\right)^2} \tag{12}$$

where:
- y_i is the actual value
- $\hat{y}_i$ is the predicted value
- $\bar{y}_i$ is the mean of actual values
- n is the number of samples

These values were determined following the prediction stage on the test data, which gives insightful picture of the average error and the degree of the generalization of the model. The trend was, that hybrid model outperformed single model in every metric since it had the lowest MAE, MSE and RMSE as well as the highest R^2 value.

4. Results and discussion

This section presents and comments the result yielded by the proposed deep neural networks–CNN, BiLSTM, TCN and the Hybrid (CNN + BiLSTM + TCN)– evaluated by using of standard regression metrics: MAE, MSE, RMSE and R^2.

In addition to reporting predictive accuracy, this section interprets how the hybrid architecture improves forecasting capability by capturing hierarchical temporal patterns. The results also demonstrate the robustness of the proposed model across different temporal splits using time-series cross-validation.

4.1. CNN model results

The Convolutional Neural Network (CNN) model was set and trained on 30 epochs to predict the succeeding values of the cryptocurrency time series through four price-based variables: Open, High, Low as well as Close. The CNN structure has used convolutional layers to extract local models and features from datasets, leading to an effective capture

Table 1
CNN model evaluation performance metrics scores.

Applied Model	MSE	MAE	RMSE	R^2
Standalone CNN	0.000201	0.008086	0.014192	0.997346

Table 2
BiLSTM model evaluation performance metrics scores.

Applied Model	MSE	MAE	RMSE	R^2
Standalone BiLSTM	0.000188	0.007894	0.013701	0.997527

Table 3
TCN model evaluation performance metrics scores.

Applied Model	MSE	MAE	RMSE	R^2
Standalone TCN	0.001026	0.018102	0.032025	0.986486

Table 4
Hybrid CNN + BiLSTM + TCN model evaluation performance metrics scores.

Applied Model	MSE	MAE	RMSE	R^2
Hybrid CNN + BiLSTM + TCN	5.0×10^{-5}	0.004438	0.007075	0.999341

of spatial correlations within the temporal series. The training loss continued to decrease during epochs, indicating the model's ability to learn complex feature representations from the input data. The validation loss showed stable coalescence, showing that the model generalizes well on invisible data without overtraining as shown in Fig. 2 .

On the test set, the CNN model has achieved a Mean Absolute Error (MAE) of 0.008086, Mean Squared Error (MSE) of 0.000201, and Root Mean Squared Error (RMSE) of 0.014192. These low error values have exceptionally indicated that the model achieved high accuracy in predicting close to the actual price values. Moreover, the coefficient of determination (R^2 score) was 0.997346, that is to say that more than 99.7% of the variance in the target variable was explained by the model,such as in Table 1 . These findings have proven that CNN is highly effective in capturing local temporal patterns and feature extraction from financial time series data. The model has displayed consistent learning patterns, smooth convergence, and excellent predictive capability, setting up a strong performance standard to which other sophisticated models such as BiLSTM, TCN and hybrid models will be compared.

However, CNN's limitation lies in its restricted ability to model long-range temporal dependencies, which are critical in financial forecasting where historical price trends influence future movements over extended periods.

4.2. BiLSTM model results

BiLSTM model has proven that it has good learning dynamics and a great ability to recognize the time patterns when used in the Bitcoin price prediction issues. As illustrated in Fig. 3,the model has achieved an outstanding confluence as the training loss varied within 4.0×10^{-4} in epoch 1 to 2.1×10^{-4} in the last epoch. The validation loss has significantly been reduced, with the initial figure being 5.8×10^{-3} and the final one being 1.87×10^{-4}, which means better generalization and stability of the model on the test data.

After the quantitative assessment, BiLSTM model reported Mean Absolute Error (MAE) score of 0.007894, Mean Squared Error (MSE) with a value of 0.000188 and a Root Mean Squared Error (RMSE) with a value of 0.013701. These low error indicators show approximate prediction accuracy without much variation between the actual values. Furthermore, the R 2 of 0.997527 indicates that the model was effective in capturing the approximately 99.75 percent of the variance in the target time series, which indicates a strong predictive analysis as shown in Table 2. The high accuracy of BiLSTM over the unidirectional LSTM has also proven that the context used in the modeling of such a sequence of finance data should be bidirectional. This bidirectional traffic allows the model to trace patterns throughout the past and future time steps, which is particularly useful when trying to identify the complex dependencies of actually how prices of coins move.

Compared to CNN, the BiLSTM model provides improved contextual understanding of price movements. However, recurrent architectures may still face limitations in efficiently modeling very long-term dependencies and may suffer from increased computational complexity.

4.3. Temporal convolutional network (TCN) model results

The Temporal Convolutional Network (TCN) model has shown a great performance in modeling the temporal sequences in Bitcoin's price data, indicating a good convergence pattern during the training. At the beginning of the training, the loss was equal to 4.0×10^{-4}. Nevertheless, by the 21st epoch it has significantly dropped to 2.5×10^{-5}. Likewise, the validation loss decreased to a minimum of 2×10^{-4}, showing successful generalization and strong learning on new data as illustrated in Fig. 4.

The TCN model has achieved a Mean Absolute Error (MAE) of 0.018102, Mean Squared Error (MSE) of 0.001026, and Root Mean Squared Error (RMSE) of 0.032025, demonstrating reasonable predictive accuracy with manageable error levels. Moreover, the coefficient of determination (R^2) reached 0.986486, indicating that the model can account for over 98.6% of the variation in the actual target values.TCN utilizes its convolutional structure to model long-term dependencies in financial time series data by detecting local temporal patterns through causal and dilated convolutions. The TCN model is capable of processing sequential data in parallel while maintaining temporal causality which makes it an effective alternative to RNN-based models, especially for time-sensitive regression problems such as cryptocurrency prediction, providing computational efficiency in parallel with reliable forecasting performance.

Although TCN effectively captures long-range memory, its performance was lower than CNN and BiLSTM when used independently, suggesting that relying solely on long-term structure may not be sufficient to model short-term volatility patterns.

4.4. Hybrid deep learning model results (CNN + BiLSTM + TCN)

The optimized hybrid model, which strategically combined Convolutional Neural Network (CNN), Bidirectional Long Short-Term Memory (BiLSTM), and Temporal Convolutional Network (TCN) branches, has shown an exceptional performance in predicting cryptocurrency time series data. The modified model structure has successfully managed to integrate local feature extraction through CNN's convolutional layers, bidirectional temporal dependency learning via BiLSTM units, and advanced temporal pattern recognition through TCN filters, generating a synergistic combination that captures multi-scale temporal dynamics. The integration of residual skip connections has contributed to the increase of the model's learning ability by enabling deeper information flow and reducing gradient vanishing problems. The training process has demonstrated noticeable convergence features, as the loss has sharply gone down during the first epoch, and an excellent stability has been achieved during the training period which lasted for 30 epochs. The validation loss has confirmed great generalization ability, the fact that emphasizes the model's effectiveness against the training abundance maintaining a predictive accuracy that surpasses invisible data. The performance evaluation results reveal outstanding accuracy, as summarized in Table 4.

These exceptional indicators demonstrate that the hybrid model explains more than 99.93% of variance in actual digital cryptocur-

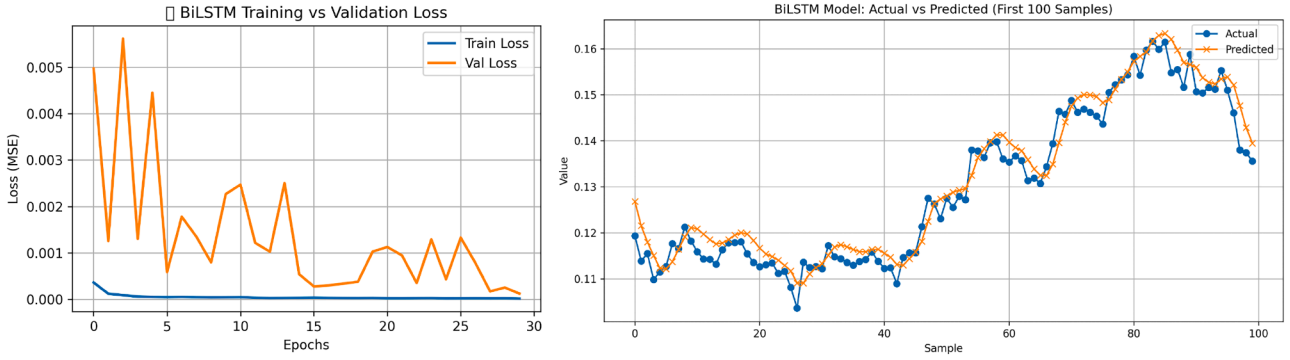


Fig. 3. Actual versus predicted Bitcoin prices using the BiLSTM model: (a) training and validation loss convergence, and (b) comparison between predicted and true prices.

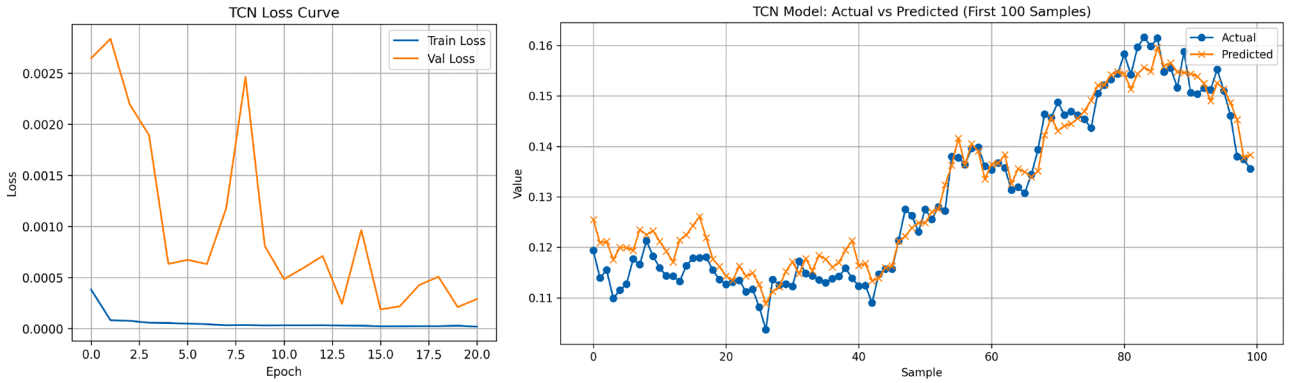


Fig. 4. Actual versus predicted Bitcoin prices using the TCN model: (a) training and validation loss convergence, and (b) comparison between predicted and true prices.

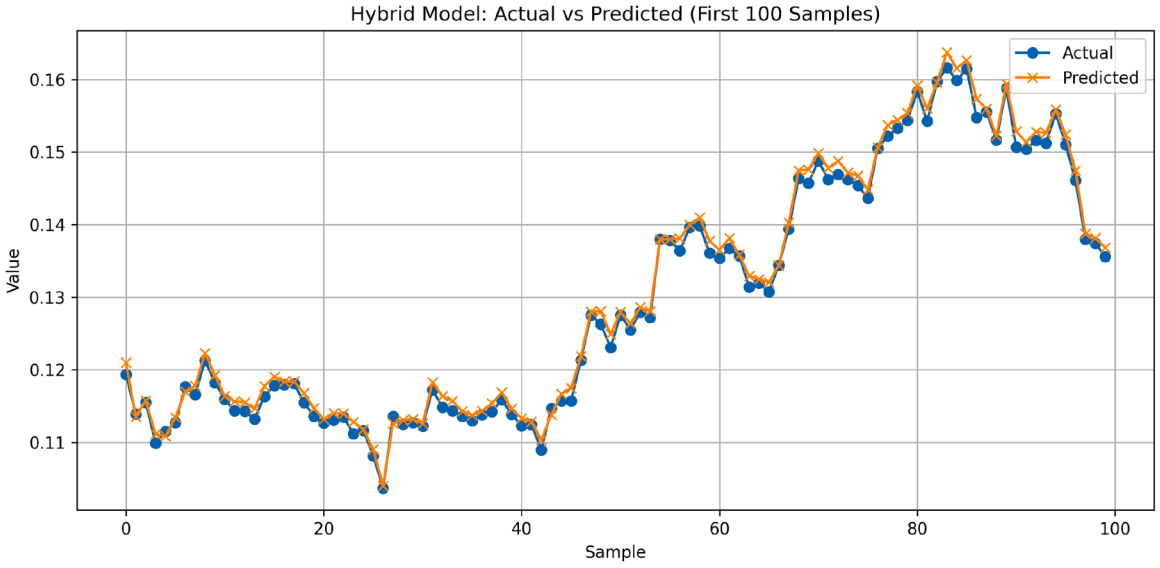
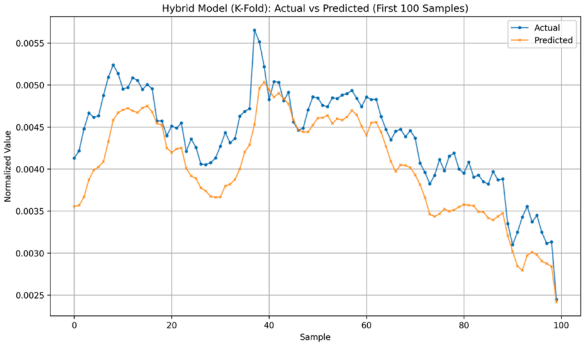
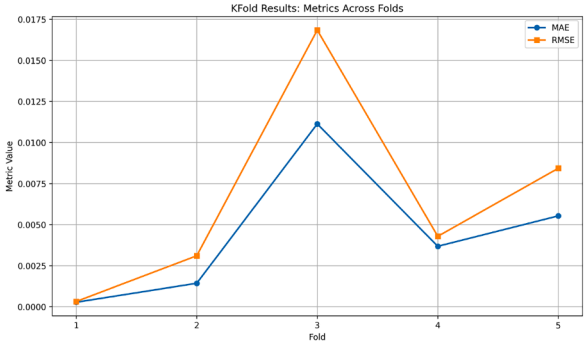


Fig. 5. Actual versus predicted Bitcoin prices using the hybrid CNN + BiLSTM + TCN model.

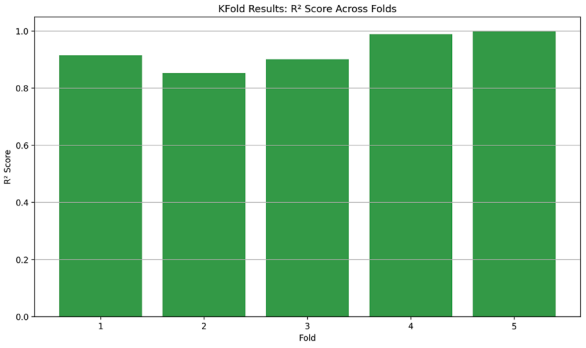
rency prices, which represents a remarkable improvement compared to the previous structure. The significant decrease in the values of both MAE (from 0.005079 to 0.004438) and RMSE (from 0.009412 to 0.007075) demonstrate enhanced precision and reliability, while the improved R^2 score (from 0.998833 to 0.999341) confirms a higher explanatory power. The sharp decrease in MSE (from 8.9×10^{-5} to 5.0×10^{-5}) reinforces the efficiency of the adopted structural enhancements. These outstanding results support the likelihood of applying the model in highly risky financial predictions scenarios, where accuracy is a decisive factor. Generally speaking, this enhanced hybrid model outperformed the previous deep-learning models in robustness and accuracy, confirming its superiority in actual financial predictions within extreme volatile markets such as cryptocurrency trading.



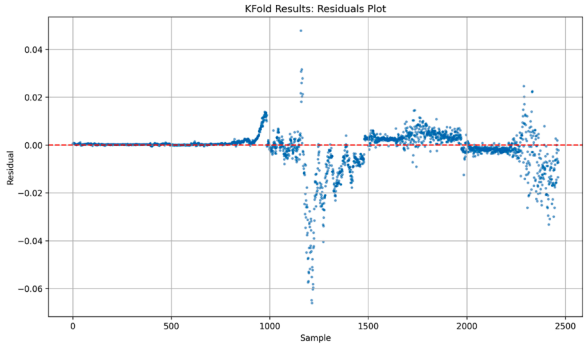
(a) Actual vs. predicted Bitcoin prices (first 100 samples).



(b) MAE and RMSE across K-folds.



(c) R^2 score distribution across folds.



(d) Residual error distribution across samples.

Fig. 6. K-fold cross-validation results of the hybrid CNN–BiLSTM–TCN model: (a) actual versus predicted prices, (b) MAE and RMSE across folds, (c) R^2 scores across folds, and (d) residual error distribution.

Table 5
Conceptual comparison of forecasting architectures.

Model Type	Strengths	Limitations
Standalone CNN	Captures local patterns and short-term trends	Limited ability to model long-term dependencies
BiLSTM	Learns bidirectional temporal relationships	Higher computational cost and longer training time
TCN	Efficient modeling of long-range dependencies	May miss localized short-term volatility signals
Transformer	Captures global dependencies via self-attention	Requires large datasets and high computational resources
Sentiment-based models	Incorporate external market signals	Sensitive to noise and data quality issues
Proposed Hybrid	Captures short-, medium-, and long-term dependencies simultaneously	Increased architectural complexity

Recent studies using Transformer-based and sentiment-integrated models have also shown promising performance; however, the proposed hybrid architecture offers a computationally efficient alternative that captures multi-scale dependencies using structured price data.

To further position the proposed model within the existing literature, a conceptual comparison with commonly used forecasting architectures is presented in Table 5.

As shown in Table 5, the proposed hybrid framework integrates the complementary strengths of CNN, BiLSTM, and TCN to capture multi-scale temporal dynamics, providing a balanced trade-off between predictive accuracy, interpretability, and computational efficiency.

4.5. K-fold cross-validation

In order to guarantee the robustness and generalizability of the proposed hybrid CNN–BiLSTM–TCN model, a comprehensive 5-fold time series cross-validation was conducted using the *TimeSeriesSplit* strategy to preserve temporal order. This validation approach is critically important for financial time series data, where temporal dependencies must be maintained during model evaluation. The K-fold validation results confirmed the model's consistent performance across different temporal segments of the dataset, with training samples progressively increasing from 496 to 2468 samples across the five folds, while maintaining 493 validation samples per fold.

Fig. 6 illustrates the outcomes of the K-fold cross-validation process. As shown in Fig. 6(a), the predicted Bitcoin prices closely follow the actual values for the first 100 samples, demonstrating strong tracking capability. Fig. 6(b) presents the evolution of MAE and RMSE across folds, indicating stable error behavior with moderate variability. The distribution of R^2 scores across folds is depicted in Fig. 6(c), highlighting consistently high explanatory power throughout different temporal splits. In addition, the residual analysis illustrated in Fig. 6(d) shows that most residuals are centered around zero, with no evident systematic bias, further validating the reliability of the proposed hybrid model.

Quantitatively, the cross-validation yielded highly promising results, with a mean MAE of 0.004410 ± 0.003820, indicating consistent

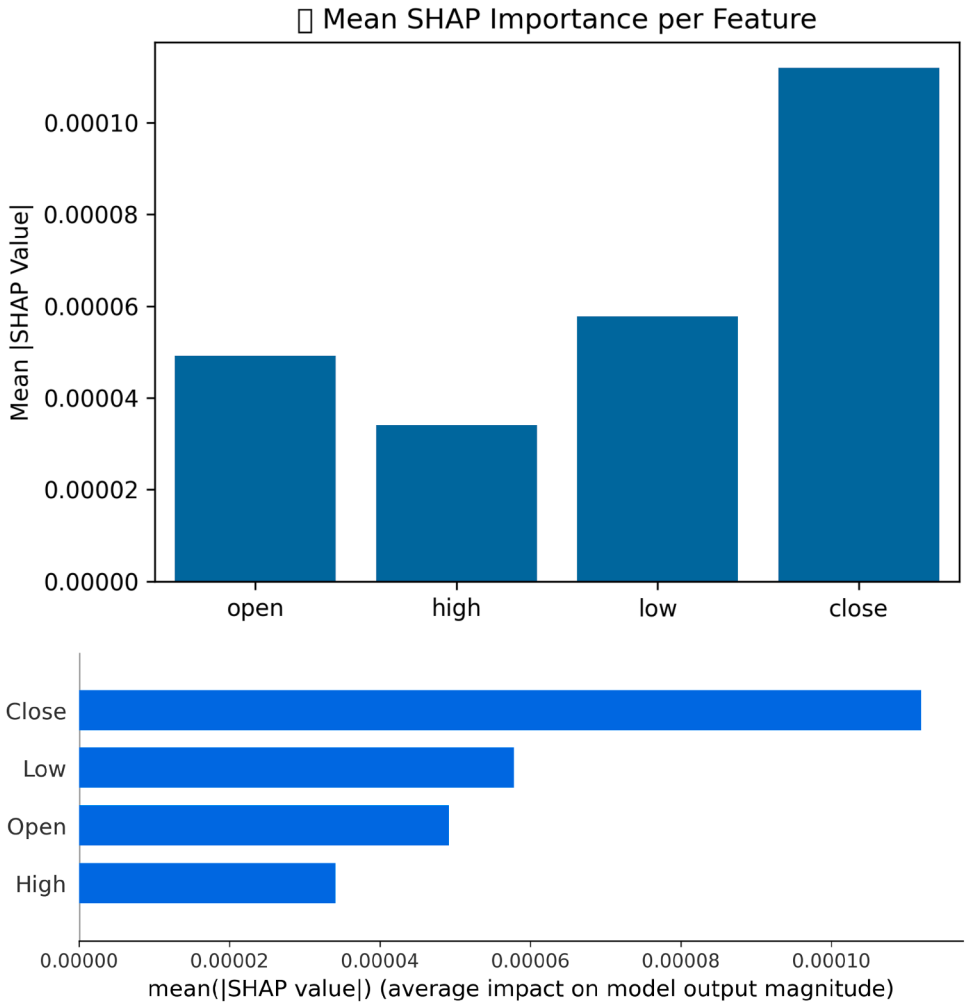


Fig. 7. SHAP-based explainability results for the hybrid CNN–BiLSTM–TCN model. The figures illustrate the global mean absolute SHAP importance of input features, highlighting the dominant influence of the closing price on model predictions.

predictive accuracy with relatively low variance across folds. The MSE reached an average of 0.000077 ± 0.000106, while the RMSE achieved 0.006597 ± 0.005750, both demonstrating excellent model stability and precision. Remarkably, the coefficient of determination (R^2) averaged 0.931095 ± 0.055506, with values ranging from 0.852039 to 0.999116, confirming the model's ability to explain over 93% of the variance in cryptocurrency price movements across different time periods.

The progressive performance improvement observed across folds, particularly in Fold 5 where the R^2 score reached 0.999116 with an MAE of 0.005533, indicates that the model benefits from increased training data, demonstrating strong scalability and learning capability. Overall, this K-fold validation framework provides statistically reliable performance estimates and confirms the effectiveness of the hybrid architecture in capturing short-term patterns through CNN layers, bidirectional temporal dependencies via BiLSTM, and long-term temporal structures using TCN components.

The consistent performance observed across folds indicates that the model maintains stability across different temporal market conditions, including both relatively stable and highly volatile periods, suggesting strong adaptability to changing cryptocurrency dynamics.

4.6. SHAP-based explainability analysis

SHAP (SHapley Additive exPlanations) was employed to interpret the predictions of the proposed hybrid deep learning model (CNN + BiLSTM + TCN) and to quantify the contribution of each input feature based on cooperative game theory. The SHAP framework provides a theoretically grounded approach for assigning importance values to features in complex, non-linear models.

In this study, a *KernelExplainer* was adopted, treating the trained hybrid model as a black box. A background dataset consisting of 50 samples was used, while SHAP values were computed for 10 samples from the test set. Given that the input tensor of the model had a shape of $(30, 4)$–corresponding to 30 time steps and four financial features (Open, High, Low, Close)–the resulting SHAP values were initially obtained in a flattened form of $(10, 120)$. These values were subsequently reshaped into $(10, 30, 4)$ to preserve the true temporal and feature-wise structure of the input data.

To extract global feature importance, the mean absolute SHAP values were averaged across both time steps and samples. Fig. 7 illustrates the resulting SHAP-based feature importance analysis. The bar plot clearly indicates that the *Close* price has the highest contribution to the model's predictions, followed by *Low*, *Open*, and *High* prices. This ranking aligns well with financial intuition, as the closing price typically encapsulates the most informative market signal.

Overall, this explainability analysis confirms that the proposed hybrid model does not behave as a black box; instead, it consistently relies on meaningful financial indicators across temporal dimensions. The SHAP-based interpretation enhances transparency and trustworthiness, which are critical requirements for deploying deep learning models in

high-risk financial forecasting applications such as cryptocurrency price prediction.

From a practical perspective, such explainability can support financial analysts and traders in understanding which price signals drive predictions, thereby enabling more informed decision-making and increasing trust in model-generated forecasts.

- **SHAP output shape:** $(10, 120)$ reshaped to $(10, 30, 4)$
- **Features analyzed:** Open, High, Low, Close
- **Explainability method:** KernelExplainer with mean absolute SHAP importance

While the proposed hybrid model demonstrates strong predictive performance, certain limitations should be acknowledged. The model is trained on Bitcoin price data and relies primarily on OHLC features, which may limit generalization to other financial assets or market conditions. Additionally, sudden external shocks such as regulatory announcements or market manipulation events may affect predictive stability. Deploying the model in real-time streaming environments may also require further optimization and computational resource management. Despite these limitations, the proposed architecture is model-agnostic and can be extended to other cryptocurrencies and financial time-series datasets with similar temporal structures. Future work will focus on incorporating external variables such as sentiment indicators, macroeconomic factors, and multi-asset datasets to further enhance robustness and generalization.

5. Conclusion

In this study, a hybrid deep model with significant and interpretable performance is introduced to the task of giving price predictions in cryptocurrencies and the model involves CNN, BiLSTM and TCN architecture.

The proposed model was trained on multivariate input features such as Open, High, Low, and Close prices over historical time windows so that it can learn both long-term dependencies and local temporal trends. The outcomes of the experiments validated that the performance of this hybrid method is extremely high when compared to individual models in terms of predictive accuracy.

The hybrid model had a Mean Absolute Error (MAE) of 0.004438, Mean Squared Error (MSE) of 5.0×10^{-5}, Root Mean Squared Error (RMSE) of 0.007075, and an R^2 score of 0.999341 on the test set which is highly accurate representing excellent generalizability.

These results demonstrate the effectiveness of combining CNN for capturing short-term patterns, BiLSTM for learning bidirectional temporal relationships, and TCN for modeling long-range dependencies within financial time-series data.

Besides high performance, the interpretability of the model was increased by the implementation of SHAP (SHapley Additive Explanations), adding a layer of transparency to the model's decision-making process. SHAP demonstrated the influence of features across time steps, allowing the relative importance of input variables to be identified. The graphical illustrations confirmed that the model consistently relies on important price characteristics such as closing and high prices, emphasizing its effectiveness in practical forecasting scenarios.

Although recent studies using Transformer-based and sentiment-integrated models have also demonstrated promising performance, the proposed hybrid architecture provides a computationally efficient and interpretable alternative that effectively captures multi-scale temporal dependencies using structured price-based features.

The combination of strong predictive performance and interpretability therefore positions the proposed hybrid model as a powerful tool for supporting real-time financial decision-making and for advancing research on cryptocurrency price dynamics Fig. 5 and Table 3.

CRediT authorship contribution statement

Rachid Bourday: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Ali Zaaouat:** Visualization, Validation, Formal analysis, Data curation, Conceptualization; **Issam Aattouchi:** Supervision, Validation, Writing – review & editing, Methodology; **Mounir Ait Kerroum:** Supervision, Project administration, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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